

UNIX / Linux — Final Exam

Subject: Part Aa

Duration: 2 hours | 15 MCQ (15 pts) + 2 Exercises (5 pts) | +1 correct / -0.25 wrong / 0 no answer

NAME: _____

GROUP: _____

 Use of phones, documents, or any form of cheating is strictly prohibited. Any student caught cheating will receive a grade of 0.

Part Aa — Multiple Choice Questions (15 pts)

Write A, B, C or D in the Ans column. +1 pt correct | -0.25 pt wrong | 0 pt no answer

Question	A	B	C	D	Ans
Q1. What does 'chmod 644 file' allow?	<code>r-x-w--w-</code>	<code>rw-rw-rw-</code>	<code>r--r--r--</code>	<code>rw-r--r--</code>	
Q2. Which command changes the file owner?	<code>chown</code>	<code>chgrp</code>	<code>usermod</code>	<code>chmod</code>	
Q3. What is the purpose of & at the end of a command?	Run in background	Increase priority	Terminate process	Send output to file	
Q4. Send report.txt to /home/user/docs/ on remote server:	<code>rsync report.txt thomas@234.322.234.10</code>	<code>ssh report.txt thomas@234.322.234.10</code>	<code>scp report.txt thomas@234.322.234.10:/home/user/docs/</code>	<code>cp report.txt thomas@234.322.234.10 remote</code>	
Q5. Which variable represents the number of arguments passed to a script?	<code>\$0</code>	<code>\$@</code>	<code>\$?</code>	<code>\$#</code>	
Q6. Connect to remote server 234.322.234.10 as thomas:	<code>scp thomas@234.322.234.10</code>	<code>login thomas@234.322.234.10</code>	<code>ssh thomas@234.322.234.10</code>	<code>ssh 234.322.234.10 thomas</code>	
Q7. What does 'export VAR=value' do?	Create alias	Delete var	Make var global	Run command	
Q8. <code>umask 022</code> → new file permission?	<code>644</code>	<code>622</code>	<code>022</code>	<code>044</code>	
Q9. Which command lists all network interfaces?	<code>ifconfig</code>	<code>netstat</code>	<code>route</code>	<code>ip a</code>	
Q10. Difference between 'export PATH=\$PATH:/opt' and 'PATH=\$PATH:/opt'?	export makes it visible to child processes	PATH= is permanent on disk	export deletes the previous PATH	Same behavior in all cases	

Question	A	B	C	D	Ans
Q11. What does 'kill -15 PID' do?	Graceful terminate	Restart process	Force kill	Pause process	
Q12. Which command lists shell background jobs?	jobs	top	ps	bg	
Q13. Which command shows running processes in real-time?	ps	bg	jobs	top	
Q14. Which command creates a group named profs?	mkgroup profs	useradd profs	groupadd profs	addgrp profs	
Q15. rsync to copy project folder from remote to /home/user/docs/:	rsync -av project thomas@234.322.234.10:/home /user/docs/	rsync -av thomas@234.322.234.10 :project /home/user/docs/	rsync -rv thomas@234.322.234.10 project /home/user/docs/	rsync project local	

Part B — Exercises (5 pts)

Exercise 1 — (2 pts)

Write a script max.sh that takes two integers as arguments and prints the maximum.

Expected usage:

```
$ ./max.sh 7 12
12
```

Answer:

Exercise 2 — (3 pts)

Write a Bash script that defines a **function named backup_txt_files**.

The function must:

1. Take one argument: the name of a destination folder.
2. Check if the argument is provided. If not, display a usage message and stop.
3. Check if the destination folder exists. If it does not exist, create it.
4. Copy all .txt files into the destination folder.
5. Compress the folder into a .tar.gz archive with the same name as the folder.

Example usage: backup_txt_files my_folder

Answer:

UNIX / Linux — Final Exam

Subject: Part Ab

Duration: 2 hours | 15 MCQ (15 pts) + 2 Exercises (5 pts) | +1 correct / -0.25 wrong / 0 no answer

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Part Ab — Multiple Choice Questions (15 pts)

Write A, B, C or D in the Ans column. +1 pt correct | -0.25 pt wrong | 0 pt no answer

Question	A	B	C	D	Ans
Q1. Which command lists shell background jobs?	<code>jobs</code>	<code>ps</code>	<code>top</code>	<code>bg</code>	
Q2. Which command changes the file owner?	<code>chmod</code>	<code>chgrp</code>	<code>usermod</code>	<code>chown</code>	
Q3. What does 'chmod 644 file' allow?	<code>rw-r--r--</code>	<code>r-x-w--w-</code>	<code>rwXrw-rw-</code>	<code>r--r--r--</code>	
Q4. What does 'export VAR=value' do?	Make var global	Create alias	Delete var	Run command	
Q5. Difference between 'export PATH=\$PATH:/opt' and 'PATH=\$PATH:/opt'?	export makes it visible to child processes	PATH= is permanent on disk	Same behavior in all cases	export deletes the previous PATH	
Q6. rsync to copy project folder from remote to /home/user/docs/:	<code>rsync -rv thomas@234.322.234.10 project /home/user/docs/</code>	<code>rsync -av project thomas@234.322.234.10:/home/user/docs/</code>	<code>rsync project local</code>	<code>rsync -av thomas@234.322.234.10 :project /home/user/docs/</code>	
Q7. Which variable represents the number of arguments passed to a script?	<code>\$0</code>	<code>\$@</code>	<code>\$#</code>	<code>\$?</code>	
Q8. umask 022 → new file permission?	622	644	022	044	
Q9. Send report.txt to /home/user/docs/ on remote server:	<code>rsync report.txt thomas@234.322.234.10</code>	<code>ssh report.txt thomas@234.322.234.10</code>	<code>scp report.txt thomas@234.322.234.10:/home/user/docs/</code>	<code>cp report.txt thomas@234.322.234.10 remote</code>	
Q10. Connect to remote server 234.322.234.10 as thomas:	<code>scp thomas@234.322.234.10</code>	<code>login thomas@234.322.234.10</code>	<code>ssh 234.322.234.10 thomas</code>	<code>ssh thomas@234.322.234.10</code>	

Question	A	B	C	D	Ans
Q11. What does 'kill -15 PID' do?	Force kill	Restart process	Pause process	Graceful terminate	
Q12. What is the purpose of & at the end of a command?	Run in background	Terminate process	Increase priority	Send output to file	
Q13. Which command creates a group named profs?	groupadd profs	mkgroup profs	addgrp profs	useradd profs	
Q14. Which command shows running processes in real-time?	ps	jobs	bg	top	
Q15. Which command lists all network interfaces?	ip a	netstat	route	ifconfig	

Part B — Exercises (5 pts)

Exercise 1 — (2 pts)

Write a script max.sh that takes two integers as arguments and prints the maximum.

Expected usage:

```
$ ./max.sh 7 12
12
```

Answer:

Exercise 2 — (3 pts)

Write a Bash script that defines a **function named backup_txt_files**.

The function must:

1. Take one argument: the name of a destination folder.
2. Check if the argument is provided. If not, display a usage message and stop.
3. Check if the destination folder exists. If it does not exist, create it.
4. Copy all .txt files into the destination folder.
5. Compress the folder into a .tar.gz archive with the same name as the folder.

Example usage: backup_txt_files my_folder

Answer:

UNIX / Linux — Final Exam

Subject: Part Ac

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Part Ac — Multiple Choice Questions (15 pts)

Write A, B, C or D in the Ans column. +1 pt correct | -0.25 pt wrong | 0 pt no answer

Question	A	B	C	D	Ans
Q1. Which variable represents the number of arguments passed to a script?	\$#	\$0	\$@	\$?	
Q2. Which command creates a group named profs?	addgrp profs	groupadd profs	useradd profs	mkgroup profs	
Q3. Which command lists shell background jobs?	top	ps	jobs	bg	
Q4. What does 'kill -15 PID' do?	Pause process	Restart process	Graceful terminate	Force kill	
Q5. Which command changes the file owner?	chown	usermod	chgrp	chmod	
Q6. Difference between 'export PATH=\$PATH:/opt' and 'PATH=\$PATH:/opt'?	export deletes the previous PATH	export makes it visible to child processes	Same behavior in all cases	PATH= is permanent on disk	
Q7. umask 022 → new file permission?	644	622	022	044	
Q8. Connect to remote server 234.322.234.10 as thomas:	ssh 234.322.234.10 thomas	scp thomas@234.322.234.10	login thomas@234.322.234.10	ssh thomas@234.322.234.10	
Q9. What does 'export VAR=value' do?	Delete var	Run command	Create alias	Make var global	
Q10. Which command shows running processes in real-time?	ps	top	bg	jobs	

Question	A	B	C	D	Ans
Q11. Send report.txt to /home/user/docs/ on remote server:	<pre>scp report.txt thomas@234.322.234.10:/home /user/docs/</pre>	<pre>rsync report.txt thomas@234.322.234.10</pre>	<pre>ssh report.txt thomas@234.322. 234.10</pre>	<pre>cp report.txt thomas@234.322.234.10 remote</pre>	
Q12. rsync to copy project folder from remote to /home/user/docs/:	<pre>rsync -rv thomas@234.322.234.10 project /home/user/docs/</pre>	<pre>rsync -av project thomas@234.322.234.10:/home /user/docs/</pre>	<pre>rsync project local</pre>	<pre>rsync -av thomas@234.322.234.10 :project /home/user/docs/</pre>	
Q13. What is the purpose of & at the end of a command?	Send output to file	Terminate process	Increase priority	Run in background	
Q14. What does 'chmod 644 file' allow?	<pre>r-x-w--w-</pre>	<pre>rw-rw-rw-</pre>	<pre>r--r--r--</pre>	<pre>rw-r--r--</pre>	
Q15. Which command lists all network interfaces?	<pre>ip a</pre>	<pre>ifconfig</pre>	<pre>netstat</pre>	<pre>route</pre>	

Part B — Exercises (5 pts)

Exercise 1 — (2 pts)

Write a script max.sh that takes two integers as arguments and prints the maximum.

Expected usage:

```
$ ./max.sh 7 12
12
```

Answer:

Exercise 2 — (3 pts)

Write a Bash script that defines a **function named backup_txt_files**.

The function must:

1. Take one argument: the name of a destination folder.
2. Check if the argument is provided. If not, display a usage message and stop.
3. Check if the destination folder exists. If it does not exist, create it.
4. Copy all .txt files into the destination folder.
5. Compress the folder into a .tar.gz archive with the same name as the folder.

Example usage: backup_txt_files my_folder

Answer:

UNIX / Linux — Final Exam

Subject: Part Ad

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Part Ad — Multiple Choice Questions (15 pts)

Write A, B, C or D in the Ans column. +1 pt correct | -0.25 pt wrong | 0 pt no answer

Question	A	B	C	D	Ans
Q1. Which variable represents the number of arguments passed to a script?	\$0	\$?	\$#	\$@	
Q2. Which command creates a group named profs?	mkgroup profs	useradd profs	addgrp profs	groupadd profs	
Q3. Difference between 'export PATH=\$PATH:/opt' and 'PATH=\$PATH:/opt'?	PATH= is permanent on disk	export deletes the previous PATH	export makes it visible to child processes	Same behavior in all cases	
Q4. What is the purpose of & at the end of a command?	Terminate process	Increase priority	Run in background	Send output to file	
Q5. Which command changes the file owner?	chown	chmod	chgrp	usermod	
Q6. What does 'export VAR=value' do?	Make var global	Run command	Create alias	Delete var	
Q7. Which command lists shell background jobs?	jobs	top	bg	ps	
Q8. Send report.txt to /home/user/docs/ on remote server:	ssh report.txt thomas@234.322.234.10	cp report.txt thomas@234.322.234.10 remote	scp report.txt thomas@234.322.234.10:/home/user/docs/	rsync report.txt thomas@234.322.234.10	
Q9. What does 'kill -15 PID' do?	Force kill	Pause process	Graceful terminate	Restart process	
Q10. Which command shows running processes in real-time?	ps	bg	jobs	top	
Q11. What does 'chmod 644 file' allow?	rw-r--r--	r-x-w--w-	rwXrW-rW-	r--r--r--	

Question	A	B	C	D	Ans
Q12. umask 022 → new file permission?	622	644	022	044	
Q13. rsync to copy project folder from remote to /home/user/docs/:	rsync project local	rsync -rv thomas@234.322.234.10 project /home/user/docs/	rsync -av thomas@234.322.234.10:project /home/user/docs/	rsync -av project thomas@234.322.234.10:/home/user/docs/	
Q14. Connect to remote server 234.322.234.10 as thomas:	scp thomas@234.322.234.10	login thomas@234.322.234.10	ssh 234.322.234.10 thomas	ssh thomas@234.322.234.10	
Q15. Which command lists all network interfaces?	ip a	route	netstat	ifconfig	

Part B — Exercises (5 pts)

Exercise 1 — (2 pts)

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Expected usage:

```
$ ./max.sh 7 12
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```

Answer:

Exercise 2 — (3 pts)

Write a Bash script that defines a function named **backup_txt_files**.

The function must:

1. Take one argument: the name of a destination folder.
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Example usage: backup_txt_files my_folder

Answer: